PROJECT SYNOPSIS

RECO CHALK

ECOFRIENDLY MULTIPURPOSE

DUSTER AND CHALK RECLAIMER



SJR PUBLIC SCHOOL

GRADE 9 , RACHITHA N

PROBLEM

In classrooms, a significant amount of chalk powder gets wasted while cleaning the black board . This, not only contributes to dust pollution , affecting health, but also leads to the wastage of usable chalk material. Additionally, broken chalk pieces are often discarded and new chalk is frequently needed. Traditional, dusters offer no solution to these problems.

PROJECT IDEA

RECO CHALK – is a multipurpose, ecofriendly duster that not only cleans the board effectively, but also collects chalk powder, stores broken chalk pieces, and has a compartment to keep full chalk pieces inside. The collected chalk powder and broken chalk pieces can be reused to make new chalk sticks, promoting recycling and reducing waste.

MATERIALS

Cardboard or hard paper

Glue / Adhesive

Rolling material [here, paint roller]

Sticks for support [here, iron nails]

Colour papers [for decoration]

Brushes

New chalk sticks

Chalk dust and broken chalk pieces

Mould and water [for making new chalk]

STEPS

1. Use a piece of hard paper and insert the iron nails by holes such that, 2 in each row [here, the paint roller should come in contact to each other] and how much ever in columns. [here, the roller should not come in contact to each other]



2. Now insert the paint rollers into it.
3. See that the front roller slides the back one and has space with the side one.
4. Now, on the other hand, create a box with 5 sides.
5. Now, through some mechanism, make the working part of the model to slide inside the box.
6. Just below the working part, add another sliding part such that , the dust collected should fall on it.



7. The working part thus can be adjusted, so, insert some brushes [here sponge] behind the working part, hence, the dust collected through the model can be separated by the brushes and the dust can fall down.



8. Check whether the working parts [rollers] are a little front to come in contact with the board while erasing the board.
9. Now, half of our model is completed, the remaining part is the chalk holder.
10. Create a box such that, it can be used as a handle to our model.
11. See through it that, it can store new chalk sticks.



12. Now, insert an opening through it.
13. Our model is ready. Check whether it works or not.

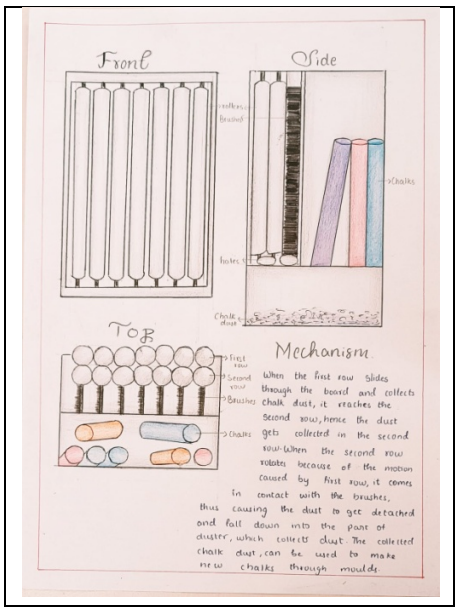
14. On the sides, we can attach sponge, so that the model works more efficiently [additional].
15. Now, decorate the model.



REUSING CHALK

1. Turn the broken chalk pieces into powder.
2. Now mix the chalk dust with water.
3. Pour into moulds and allow to dry to form new chalk sticks.

SO, OUR MODEL PLAN IS LIKE :



PROBLEM STATEMENT

In classrooms, a considerable amount of chalk powder is wasted while cleaning the blackboard. This not only contributes to dust pollution but also poses serious health risks such as – Dust allergies, Respiratory problems and discomfort for both teachers and students. In addition, usable chalk material is lost, as broken chalk pieces are often discarded, leading to frequent purchases of new chalk. Traditional dusters fail to address these problems, creating the need for an innovative, ecofriendly and sustainable solution.

CONCLUSION

RECO CHALK is an innovative, multipurpose solution that addresses classroom cleanliness, and resource reuse. It reduces chalk dust, provides convenient chalk storage and encourages eco-consious behavior in schools.With simple materials and easy assembly, it is a cost-effective and sustainable choice for every classrooms.